

本科毕业设计（论文）中期检查报告

专业：

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报告题目		The study on skin disease classification method based on deep learning			
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Research on Deep Learning-Based Skin Disease Image Classification Method

1. Task and Progress of Thesis Work

1.1 Overview of Tasks

(1) Study on skin disease classification method based on deep learning

Since starting this project, the primary focus has been transitioning from theoretical research to building a working deep learning model for skin disease classification. The research progress includes a thorough literature review, the selection of the Derm7pt dataset, and the successful design of a Dual-Branch ResNet50 neural network. Instead of just looking at one photo, this model is designed to analyze two different types of images for every patient—much like how a doctor uses both their eyes and a special magnifying tool to make a diagnosis. All major milestones for this period, including data organization and initial model training, have been completed as planned.

1.2 Progress of Thesis Work

1.2.1 Dataset Preparation

The project utilizes the Derm7pt dataset, which provides a unique collection of 1,011 paired image samples featuring both clinical and dermoscopic views of skin lesions. The preparation process involves cleaning the data and resizing all images to a standard 224x224 pixels to ensure the model can process them. To address the significant class imbalance—where some conditions like Nevus are much more common than Melanoma —weighted random sampling and stratified splits were implemented to ensure the model learns to identify rare but critical

diseases effectively.



Fig.1 Clinical vs Dermoscopy Samples

A major challenge in this work is that different skin diseases often look very similar to each other, which can easily confuse a computer model. For example, Melanoma (MEL), which is a life-threatening cancer, can appear almost identical to a Melanocytic Nevus (NV), or common mole, in a standard clinical photo.

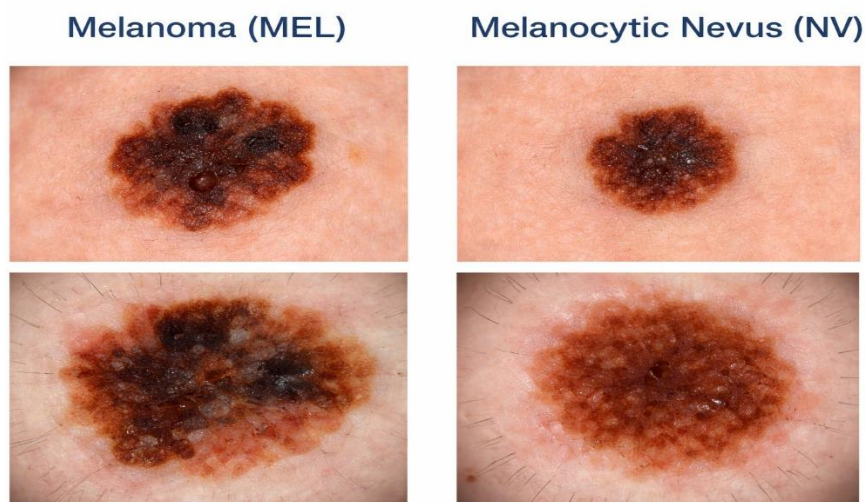


Fig.2 Melanoma vs Melanocytic Nevus Comparison

Both conditions often show up as dark, asymmetrical patches with uneven borders. This is precisely why having paired images is so important; the dermoscopic view can reveal subtle textures and underlying patterns that are invisible to the naked eye, helping the model accurately tell these similar-looking diseases apart.

1.2.2 Backbone Network Design and Implementation

The primary backbone selected for this research is the ResNet50 architecture, utilized through a transfer learning approach. This deep residual network is highly effective at capturing complex visual patterns from medical images. By using a pre-trained version of ResNet50, the model leverages existing knowledge of shapes and textures, which are then fine-tuned specifically for dermatological features.

1.2.3 Design of Frame of Network

The overall framework is a Dual-Branch Multimodal Network. This architecture consists of two parallel ResNet50 branches: one dedicated to extracting features from standard clinical "naked-eye" photos and the other focusing on deep-tissue patterns in dermoscopic images. These features are then combined and compressed using a 1×1 convolutional layer before the final classification.

1.2.4 First Results

Preliminary experimental runs reached a best validation accuracy of approximately 72.37%. The model successfully learned to distinguish between highly similar lesion types, such as Melanoma and benign Nevus, by effectively integrating visual cues from both image branches. Early Stopping was successfully implemented during training to prevent the model from overfitting to the data.

1.3 Implementation of the ResNet-50 Baseline Structure

The classification model uses a sophisticated Dual-Branch architecture based on the ResNet50 framework. This means the model actually has two separate "brains" working at the same time: one analyzes the surface-level clinical photo while the other analyzes the deep-tissue dermoscopic photo. These two sets of information are then combined into a single feature vector and compressed through a final layer to produce a final diagnosis. This dual-perspective approach is more powerful than standard models because it captures both the visual texture and the underlying structural patterns of skin lesions.

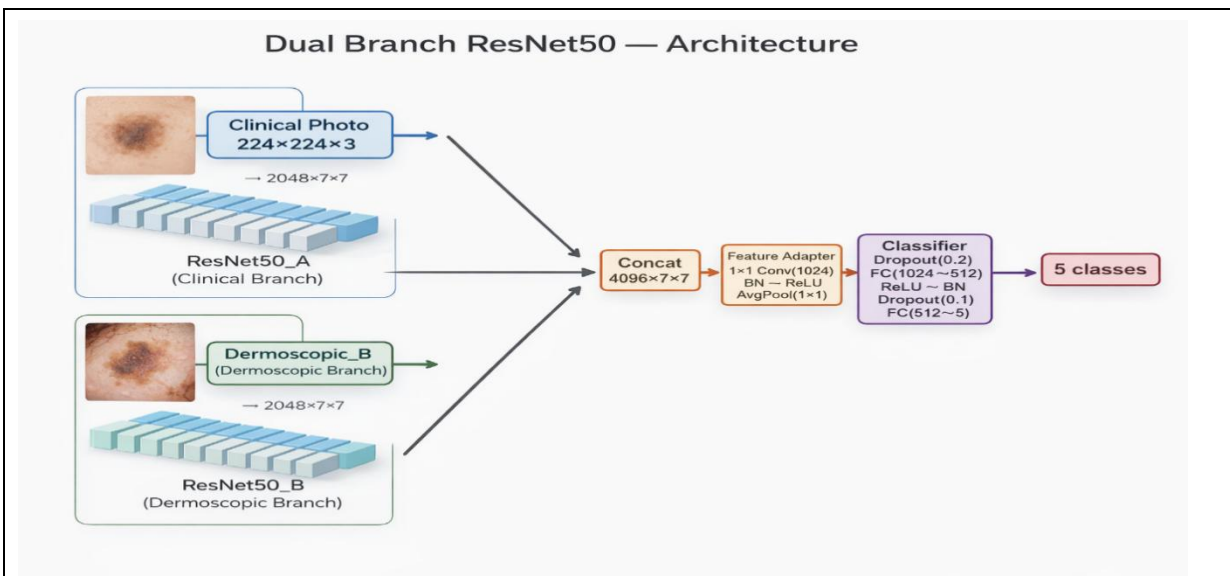


Fig.2 Dual Branch Architecture

To implement this architecture in PyTorch, a custom model class was defined to load the pre-trained ResNet-50 weights and manage the two separate image inputs. As shown in the core code below, the standard classification head of the ResNet-50 was removed using `nn.Sequential` to isolate the feature extraction layers. In the forward function, the clinical and dermoscopic images are passed through their respective branches (`branch_a` and `branch_b`). The resulting feature maps are flattened and concatenated into a single tensor before being passed into the final classifier block to output the predictions for the 5 disease classes.

```
import torch
import torch.nn as nn
import torchvision.models as models

class ResNet50Backbone(nn.Module):
    def __init__(self):
        super().__init__()
        base = models.resnet50(weights=models.ResNet50_Weights.DEFAULT)
        # Extract features without the final classification layer
        self.features = nn.Sequential(
            base.conv1, base.bn1, base.relu, base.maxpool,
            base.layer1, base.layer2, base.layer3, base.layer4
        )
    def forward(self, x):
        return self.features(x)
```

Fig.3 Feature Extractor (Backbone Code)

This class defines how the pre-trained ResNet-50 is modified to extract features without

making a final prediction.

```
class DualBranchResNet(nn.Module):
    def __init__(self, num_classes=5):
        super().__init__()
        self.resnet_clinic = ResNet50Backbone()
        self.resnet_derm = ResNet50Backbone()

        # 1x1 Conv to compress 4096 channels down to 1024
        self.compression = nn.Sequential(
            nn.Conv2d(4096, 1024, kernel_size=1),
            nn.BatchNorm2d(1024),
            nn.ReLU()
        )
        self.pool = nn.AdaptiveAvgPool2d((1, 1))
        self.classifier = nn.Sequential(
            nn.Dropout(p=0.2),
            nn.Linear(1024, num_classes)
        )

    def forward(self, img_clinic, img_derm):
        # Extract features from both branches
        feat_clinic = self.resnet_clinic(img_clinic)
        feat_derm = self.resnet_derm(img_derm)

        # Fuse and compress features
        fused = torch.cat([feat_clinic, feat_derm], dim=1)
        compressed = self.compression(fused)

        # Pool, flatten, and classify
        out = self.classifier(self.pool(compressed).flatten(1))
        return out
```

Fig. 4 The Dual-Branch Framework

This class defines how the two backbones are brought together, how the features are fused, and how the final diagnosis is made.

1.4 Preliminary Experimental Results and Analysis

The initial tests of the model have yielded promising results, achieving a validation accuracy of approximately 72.37%. The model demonstrated a strong ability to distinguish between difficult-to-diagnose conditions, such as Melanoma and benign moles, by using the combined data from both image branches. To ensure the model remains accurate and doesn't just "memorize" the training photos, Early Stopping was implemented to halt training at the optimal moment. These results confirm that the dual-branch strategy is effective, and future work will focus on adding "attention mechanisms" to further refine the model's diagnostic

precision.

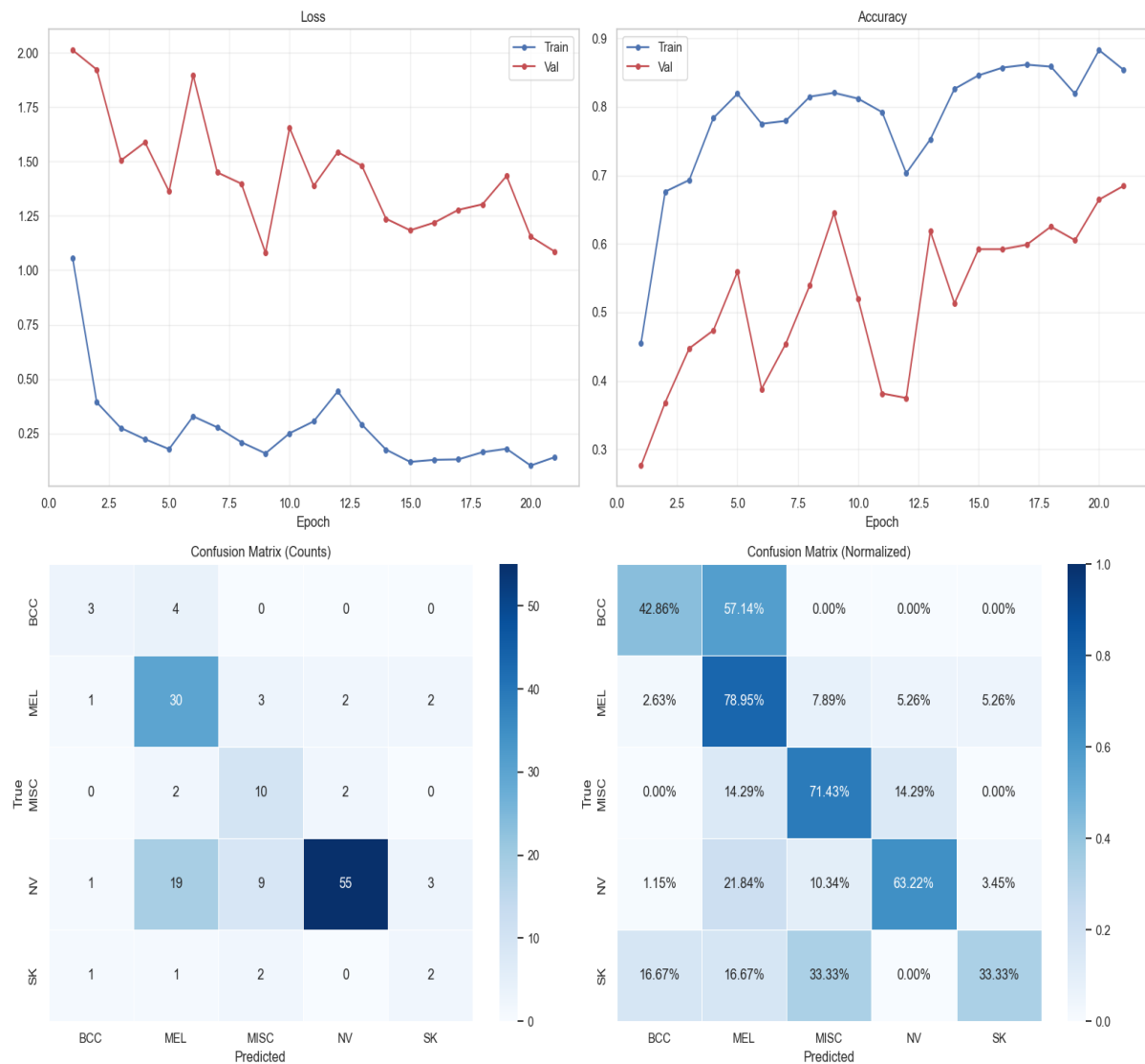


Fig.5 Training/Validation Curves

The graphs above show how the model improved over time during its training. At first, the errors (loss) drop quickly and the accuracy shoots up. This means the model is quickly learning the most obvious differences between the skin diseases. After a while, the lines start to flatten out, which shows the model has learned the main patterns and is finalizing its knowledge. You might notice some small bumps in the "validation" line; this is totally normal because many skin diseases look very similar, making the test slightly harder. To make sure the model actually learned the patterns instead of just "memorizing" the training photos (a problem called overfitting), I used Early Stopping. This automatically stopped the training at the exact moment the model performed best, ensuring the final system is accurate and reliable for real-world use.

2. Review of Relevant Literature and Materials

2.1 Advances in Deep Learning for Skin Lesion Classification

The foundation of this research is built upon the significant progress made in using Convolutional Neural Networks (CNNs) for medical image analysis. Specifically, the ResNet50 architecture is a focal point in recent literature due to its ability to extract deep, complex features while maintaining training stability through residual connections. By utilizing transfer learning, the model can adapt knowledge learned from millions of general images to the highly specific task of identifying skin diseases. This approach is particularly effective for medical tasks where specialized data is more difficult to collect than general imagery.

2.2 Multimodal Learning and the Derm7pt Dataset

Recent studies suggest that a single image often lacks the full context needed for a definitive diagnosis, leading to an increased focus on multimodal learning. The Derm7pt dataset is a critical material for this project as it provides paired sets of clinical and dermoscopic images for each case. Literature in this field indicates that clinical photos capture surface-level "naked-eye" features like lesion boundaries, while dermoscopic photos provide a magnified view of deeper pigment structures. By processing both types of images simultaneously, the model can mimic the dual-perspective approach used by expert dermatologists in clinical practice.

2.3 Addressing Data Imbalance and Model Interpretability

A recurring challenge highlighted in dermatological research is the "class imbalance" problem, where some skin conditions are much more common than others. To solve this, current methodologies recommend using weighted loss functions and weighted random sampling to ensure the model does not become biased toward the most frequent diseases. Furthermore, the literature emphasizes the importance of "explainable AI" through tools like Grad-CAM visualization. This technology allows researchers to see exactly which parts of a skin lesion the model is focusing on, ensuring that the classification is based on medically relevant patterns rather than background noise.

3. Existing Problems and Countermeasures

3.1 Small Dataset Size and Data Diversity

One of the primary challenges in this research is the relatively small size of the Derm7pt dataset, which contains only 1,011 samples from a single institution. In deep learning, a small dataset can lead to "overfitting," where the model memorizes specific images rather than learning general rules for diagnosis. To counter this, I have implemented Transfer Learning using a pre-trained ResNet50 model, allowing the system to start with existing "knowledge" of shapes and textures. I also use Data Augmentation (rotating and flipping images) and Early Stopping to ensure the model remains flexible and accurate when seeing new data.

3.2 Addressing Class Imbalance

A common problem in medical datasets is "class imbalance," where some diseases like Nevus (NV) have many more examples than rare conditions like Melanoma (MEL). If left unaddressed, the model might simply "guess" the most common disease every time to get a high score. To solve this, I have applied Weighted Random Sampling and Weighted Loss Functions in the training code. These measures force the model to pay extra attention to the rarer, more dangerous diseases during the learning process, ensuring a fair and safer diagnostic result across all categories.

3.3 Enhancing Model Interpretability

While the dual-branch model is performing well with an accuracy of 72.37%, "black box" neural networks can be difficult for doctors to trust because they don't explain why they chose a specific diagnosis. Currently, the model provides a result but does not highlight which specific part of the skin lesion was most important. As a countermeasure, the next phase of work involves implementing Grad-CAM visualization and Attention-based fusion. These tools will create "heatmaps" on the images to show exactly which spots the model is looking at, making the system much more transparent and useful for clinical review.

4. (Work Plan for the Next Stage)

4.1 Model Refinement and Robustness Testing

The next immediate step is to move beyond the initial training phase and ensure the model's results are consistent and reliable. I plan to implement K-fold cross-validation, which

involves training and testing the model on different subsets of the data multiple times. This will help confirm that the accuracy of 72.37% is stable across the entire dataset and not just a result of one specific data split. I will also continue to fine-tune the hyperparameters of the Dual-Branch ResNet50 to see if the classification of rare disease classes can be further improved.

4.2 Implementing Attention Mechanisms and Explainability

To make the model more advanced, I will work on the Attention-based feature fusion mentioned in the current proposal. This involves adding a "cross-attention" layer that helps the model automatically decide which parts of the clinical image and which parts of the dermoscopic image are most important to look at simultaneously. Furthermore, I will implement Grad-CAM visualization. This will generate "heatmaps" on the skin lesion photos, allowing me to verify that the model is focusing on medically relevant features—like irregular borders or pigment networks—rather than background noise.

4.3 Final Evaluation and Thesis Compilation

Once the model architecture is finalized with attention modules, I will conduct a comprehensive final evaluation using standard metrics such as Precision, Recall, and F1-score for all five disease classes (MEL, NV, BCC, SK, and MISC). The final stage of the project will involve documenting the entire development process, from data preprocessing to the final experimental results, into the formal thesis. I will focus on analyzing how the multimodal dual-branch approach provides a superior diagnostic tool compared to traditional single-image methods.

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Advisor's Opinion:

进度尚可,后期加快论文进度,保证论文
质量。

Signature: 杨波

Date: 2026/04/16 (YYYY/MM/DD)

The Mid-term Assessment Team:

陈建忠、 陈小锋、 杨波

Assessment of the Project (Thesis):

1. Progress: ☒ fast; ☐ normal; ☐ slow; ☐ very slow.
2. Quality: ☒ good; ☐ average; ☐ poor; ☐ very poor.
3. Conclusion: ☒ qualified; ☐ basically qualified; ☐ not qualified

(mark "√" in the corresponding box)

Assessment
Opinion

This project (thesis) has been assessed as:

- ☒ qualified;
☐ not qualified

(mark "√" in the corresponding box)

检查合格。

Leader of the Assessment Team(signature): 陈建忠

Date: 2026.04.16 YYYY/MM/DD